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Palaeomagnetism of Some Igneous Rocks of the Middle Jurassic Chon-Aike Formation from Estancia La Reconquista, Province of Santa Cruz, Argentina

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Summary

New palaeomagnetic data for the Middle Jurassic Chon-Aike Formation from 15 different geological units of rhyolitic porphyries, keratophyres, ignimbrites, pumicites and associated tuffs exposed at the Estancia La Reconquista, Province of Santa Cruz (48°10′ S, 68°50′ W), Argentina, are given.

After cleaning, all the units show normally polarized magnetic remanence and the mean of their virtual geomagnetic poles is not coincident with the mean of the virtual geomagnetic poles derived from the Puerto Deseado (Valencio & Vilas) and Camarones (Creer *et al.*) areas.

It is proposed that the most reliable palaeomagnetic pole for the Chon-Aike Formation is that given by the mean of the 54 virtual geomagnetic poles derived from all three areas, namely Puerto Deseado, Camarones and Estancia La Reconquista. It is situated at 85° S, 163° W, with $\alpha_{95} = 6°$. Comparison of the position of this palaeomagnetic pole with the mean of the African Jurassic palaeomagnetic poles suggests that the separation of South America and Africa occurred after the Middle Jurassic.

1. Introduction

Palaeomagnetic studies of Jurassic rocks of South America are of basic importance in defining the beginning of the fragmentation of the South America–Africa block. This was noted by Valencio & Vilas (1969) who, on the basis of the available palaeomagnetic data, indicated the importance of knowing the Jurassic polar position for South America. These authors started the palaeomagnetic studies of lavas of the Chon-Aike Formation of Middle Jurassic age from Argentina. The first Jurassic palaeomagnetic pole for South America was computed in the study carried out with igneous rocks of this formation from Puerto Deseado, Province of Santa Cruz (47°47′ S, 66° W). Comparison of the position of this pole with those of the Jurassic palaeomagnetic poles for Africa (Briden 1967) suggested that the opening of the South Atlantic occurred before the Middle Jurassic (Valencio & Vilas 1970). Later on Valencio (1970a, b, c) indicated the necessity of verifying the position of this pole; local reasons (tectonics, secular variation) might affect it. In particular, the interpretation concerning the time of opening of the South Atlantic conflicts with

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sea-floor spreading data (Maxwell *et al.* 1970). Following that suggestion, B. J. J. Embleton, during his visit to Argentina in 1970, collected 70 oriented hand samples of lavas of the Chon-Aike Formation from the exposures present near Camarones, Province of Chubut (44° 45′ S, 65° 65′ W), and at the same time the author started the study of other sections of the same formation cropping out at Estancia La Reconquista, Province of Santa Cruz (48° 10′ S, 68° 50′ W).

The palaeomagnetic data for the Chon-Aike Formation exposures in Puerto Deseado were obtained from 70 hand samples collected at 22 different sites; most of these collecting sites are situated at different stratigraphic levels over a distance of about 20 km (Valencio & Vilas 1970). Samples from 21 of these sites showed normal polarity; samples collected from the remainder of the sites showed reversed polarity. On the basis of this evidence the pole computed for the mean cleaned remanance of 66 of the hand samples was considered as a reliable palaeomagnetic pole (Valencio & Vilas 1970).

Samples from the Chon-Aike Formation exposures in Camarones were collected by Embleton at 17 different sites; most of these sites are situated at different stratigraphic levels. Samples from nine of these sites showed reversed polarity; samples collected from the remainder of the sites showed normal polarity. The palaeomagnetic study of these samples was carried out by Creer, Mitchell & Abou Deeb (1972) who presented a new pole for the Chon-Aike Formation obtained as the mean of the 17 virtual geomagnetic poles computed for the collecting sites. They suggested that the earlier palaeomagnetic pole for the Chon-Aike Formation (Valencio & Vilas 1970) is a virtual geomagnetic pole and proposed the pole they obtained as the more reliable palaeomagnetic pole for this formation.

In this paper the results of the palaeomagnetic study of the Chon-Aike exposures from Estancia La Reconquista are given. A revised interpretation of all available palaeomagnetic data of the Chon-Aike igneous rocks collected in Puerto Deseado, Camarones and Estancia La Reconquista is also given. This new interpretation supersedes the earlier interpretation made by Valencio & Vilas (1970).

2. Available data

2.1 *Geological evidence*

Collection of hand samples of the Chon-Aike Formation from Estancia La Reconquista, Province of Santa Cruz, Argentina, was carried out on the basis of the geologic study done by De Giusto (1956), and a schematic geologic map of the area is shown in the left hand of Fig. 1. In this figure the strike and dip of the Chon-Aike lavas (De Giusto 1956) and the sampling sites are also shown. A section of more than a thousand meters of the Chon-Aike Formation outcrops in this zone. The exposures consists of rhyolite porphyries, keratophyres, ignimbrites, pumicites and associated tuffs. De Giusto (1956) has suggested a Middle Jurassic age for the Chon-Aike Formation; radiometric studies (K–Ar) carried out on igneous rocks of this formation support this age (Cazeneuve 1965; Creer *et al.* 1972).

In the right hand of Fig. 1 the geographical positions of the three zones (Puerto Deseado, Camarones and Estancia La Reconquista), where oriented hand samples of the Chon-Aike Formation were collected, are shown.

2.2 *The palaeomagnetic data*

Estancia La Reconquista area. Fifty-six hand samples of igneous rock from the Chon-Aike Formation were collected at the 15 sites shown in Fig. 1. These sites are situated at different stratigraphic levels. The orientation of samples was carried out using both sun and magnetic compasses. Three disks 2·5 cm in diameter and 1 cm thick were cut from each hand sample. Natural remanent magnetization of these

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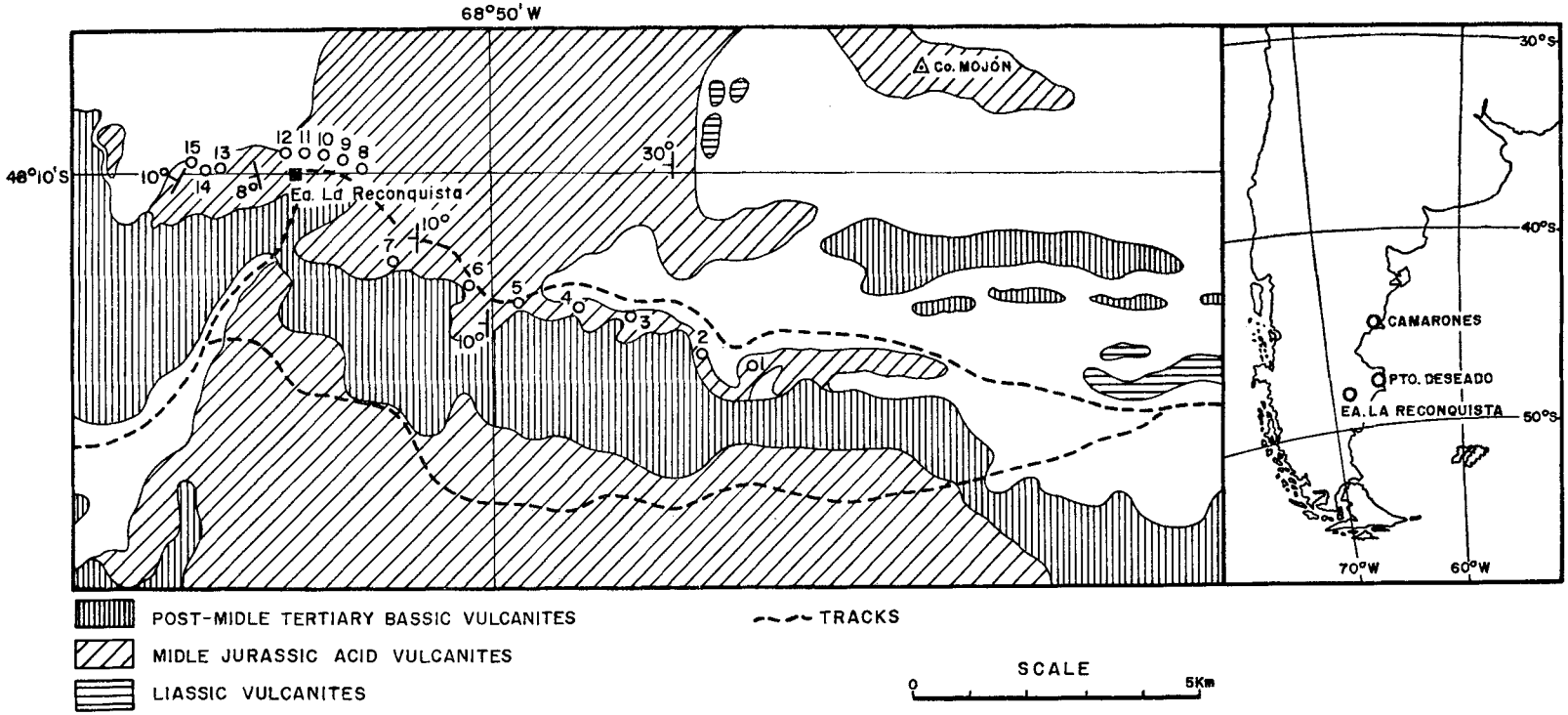


FIG. 1. Schematic geological map of the Estancia La Reconquista zone, Province of Santa Cruz, Argentina, (De Giusto 1956). Sampling sites are indicated by open circles (left). Geographical positions of zones where igneous rocks of the Chon-Aike Formation were collected: Puerto Deseado (Valencio & Vilas 1970), Estancia La Reconquista, Province of Santa Cruz, and Camarones (Creer *et al.* 1972), Province of Chubut, República Argentina (right).

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disks was measured using a spinner magnetometer. Detailed AC cleaning was used to determine the stability of the natural remanent magnetization of the samples. Two disks of each sample were subjected to AC demagnetizing fields increasing in steps of 50 Oe up to 300 Oe, in two independent processes. In this way the best AC demagnetizing field isolating the primary remanent magnetization of each sample was determined. The criterion adopted for the choice of the best AC demagnetizing field strength was that the direction of the remanence of the pilot disks exhibited no further change in response to higher demagnetizing field strengths apart from anhysteretic effects. This technique allows us to test the consistency of the measurements and to confirm that the disks had not received an anhysteretic remanence while being cleaned. All the samples showed high AF coercivities; samples with the highest AF coercivities were demagnetized to about 70 per cent of the natural intensity after the step of 300 Oe without any noticeable change in the remanent direction. A third disk of each sample was subjected to thermal cleaning. All the samples gave reliable results. Briefly the characteristics of the remanence of the Chon-Aike lavas are: (i) natural remanence homogeneous and isotropic; (ii) hard remanence; (iii) Curie point of about 675 °C; and (iv) normal polarity.

After cleaning, the directions of remanence of samples from within each site fell into a compact group. The statistical method of Fisher (1953) was used to

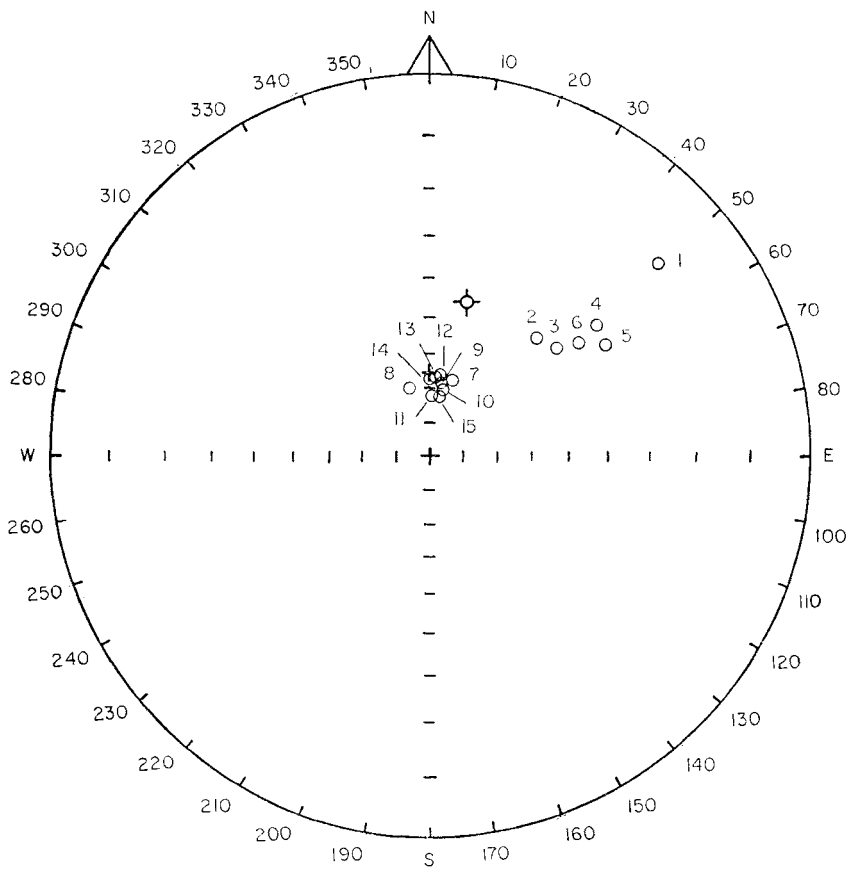


FIG. 2. Site mean directions of cleaned remanence for the Chon-Aike lavas from Estancia La Reconquista. The axial dipole normal field direction is represented by + and the present field direction by ⊹.

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Table 1

Summary of the palaeomagnetic data of the Chon–Aike igneous rocks collected in the Estancia La Reconquista zone.

Geological unit	Number of samples	Number of disks	Directions of cleaned nrm referred to the palaeo-horizontal D (°)	I (°)	α_{95} (°)	R	Polarity	Virtual geomagnetic pole Long. (East) (°)	Lat. (South) (°)	$d\psi$ (°)	$d\chi$ (°)
1	3	6	50	−14	5	5·975	*N*	174	31	2	5
2	3	6	42	−45	10	5·889	*N*	183	51	8	12
3	4	8	50	−43	7	7·875	*N*	188	45	6	9
4	4	8	52	−32	6	7·918	*N*	183	38	4	7
5	4	8	58	−33	3	7·972	*N*	189	35	2	4
6	4	8	53	−38	6	7·930	*N*	188	40	4	7
7	4	8	16	−67	7	7·889	*N*	214	80	9	12
8	5	10	343	−69	7	9·788	*N*	349	78	11	13
9	3	6	8	−68	8	5·925	*N*	236	84	12	14
10	3	6	10	−70	4	5·984	*N*	252	81	6	7
11	4	8	2	−72	6	7·918	*N*	284	80	10	11
12	4	8	7	−66	6	7·909	*N*	211	85	8	10
13	3	6	4	−67	7	6·949	*N*	226	87	9	11
14	4	8	0	−67	4	7·955	*N*	301	88	6	7
15	4	8	6	−72	2	7·986	*N*	268	81	4	4

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compute site-mean directions of remanence, giving unit weight to the direction of the remanence of each one of the disks submitted to AC cleaning (Table 1). Tectonic corrections were applied. Site-mean directions of cleaned remanence are plotted in Fig. 2; they fall into two groups: (i) those for the nine younger stratigraphic levels sampled close to Estancia La Reconquista (sites 7–15) form a compact group, the mean direction being $D = 4{\cdot}2°$ $I = -68{\cdot}9°$, $\alpha_{95} = 2{\cdot}5°$ and (ii) a group formed by the site-mean directions for the six older stratigraphic levels (sites 1–6) with the following mean direction: $D = 51°$, $I = -34$, $\alpha_{95} = 10°$. Possible explanations of this observation are: (i) the natural remanence of samples may have not been completely cleaned, (ii) the occurrence of unrecognized post-emplacement tectonic movements, and (iii) a variation in the geomagnetic field. As regards (i), it would appear that the lightly smeared position of the mean direction for site 1 (Fig. 2) might be indicative of this; however the remaining 14 site-mean directions form two circular populations, so we may rule out this possibility. With regard to (ii) structural corrections were applied according with the bedding plane data given in the detailed geological study carried out by De Giusto (1956); there is no evidence suggesting important relative movements between the two areas such as to bring the two groups into coincidence, so we may discard this possibility. As regards (iii), it may be

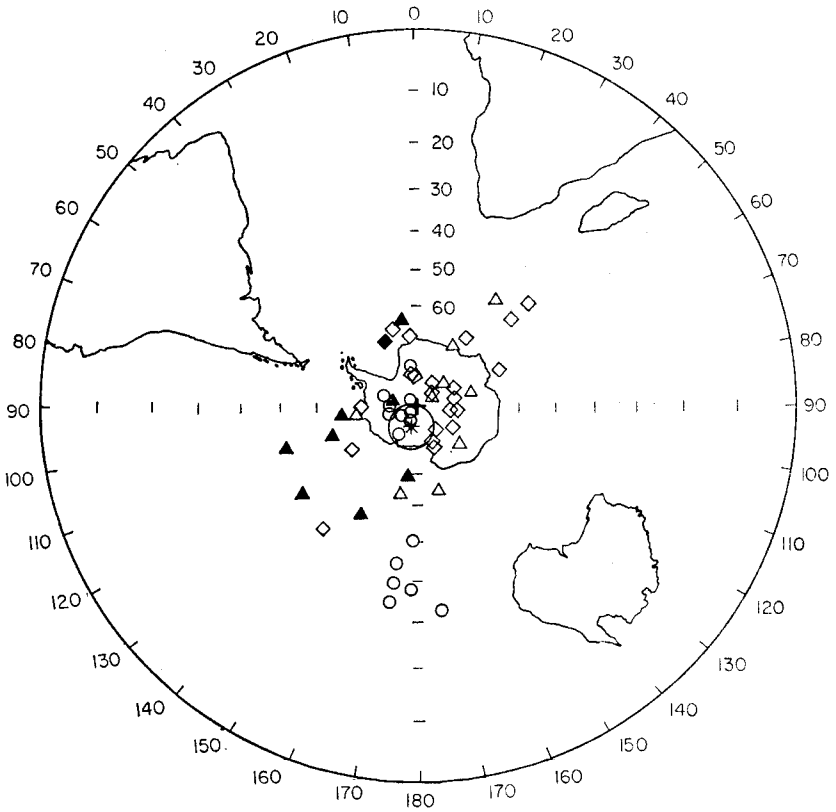


FIG. 3. Virtual geomagnetic poles for the 54 studied sites of the Chon-Aike Formation. Virtual geomagnetic poles of the Estancia La Reconquista zone (Table 1) are shown by circles, of the Puerto Deseado zone (Table 2) by diamonds and of the Camarones zone (Creer *et al* 1972) by triangles. Symbols are shaded according to the polarity of the magnetic remanence of the rocks: white symbols indicate normal polarity and black symbols reversed polarity. The mean of the 54 virtual geomagnetic poles (represented by *) and its circle of confidence are also given.

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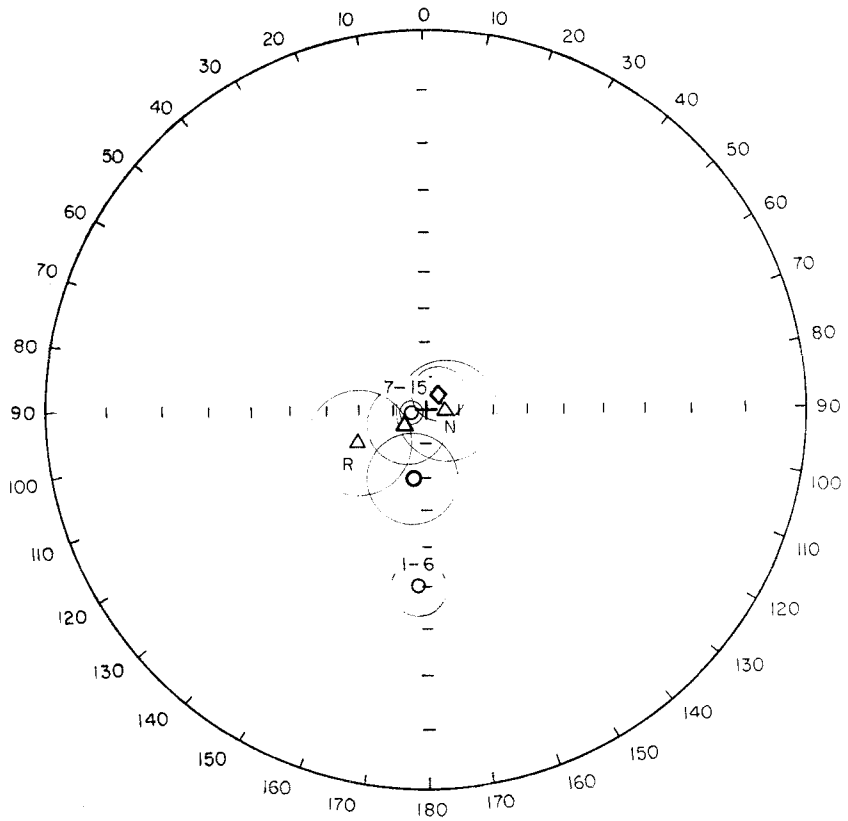


FIG. 4. Mean poles for the Chon–Aike Formation. Symbols as in Fig. 3. 1–6 and 7–15 are the poles for the older and younger units from La Reconquista. *N* and *R* are the poles for the normal and reverse polarity sites from Camarones. Thicker symbol indicates the mean pole for all the sites from one area.

produced by either (a) the secular variation, or (b) a reversal of the geomagnetic field. The difference between the site-mean directions supports the second interpretation (b). To prove this explanation it would be necessary to obtain reversed data from older units but unfortunately it was not possible to extend the sampling to units older than those of site 1 at Estancia La Reconquista. However, reversed polarity has been found in samples of the Chon-Aike Formation from Camarones (Creer *et al.* 1972) and Puerto Deseado (Valencio & Vilas 1970).

Virtual geomagnetic poles for the sampling sites of Estancia La Reconquista are presented in Table 1 and plotted in Fig. 3. The mean of these poles is given in Table 3 and shown in Fig. 4. The mean of the virtual geomagnetic poles for the younger units (sites 7–15) is situated at 85° S, 256° E with $\alpha_{95} = 4°$ and for the older units (sites 1–6) at 40° S, 184° E with $\alpha_{95} = 8°$ (Fig. 4).

Puerto Deseado area. In this sub-section a new interpretation of the palaeomagnetic data obtained by Valencio & Vilas (1970) for the Puerto Deseado exposures of the Chon-Aike Formation is given. These authors studied 70 oriented hand samples collected at 22 different sites. AC and thermal cleaning were used to determine the stability of the natural remanent magnetization of the samples. Sixty-six samples showed stable magnetic remanences. On the basis of the mean direction of the cleaned remanence of these samples, the authors computed a palaeomagnetic pole

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Table 2

Summary of the palaeomagnetic data of the Chon–Aike lavas collected in Puerto Deseado.

Site	Number of samples	Number of disks	Direction of cleaned nrm					Virtual geomagnetic poles			
			D (°)	I (°)	α_{95} (°)	R	Polarity	Long. (East) (°)	Lat. (South) (°)	$d\psi$ (°)	$d\chi$ (°)
1	4	8	15	−75	20	7·198	*N*	269	73	33	36
2	4	8	12	−58	6	7·923	*N*	161	78	6	9
3	4	8	348	−69	12	7·660	*N*	351	81	18	21
4	4	8	324	−73	10	7·759	*N*	342	66	18	19
5	2	4	328	−57	10	3·966	*N*	36	65	10	14
6	2	4	339	−46	16	3·909	*N*	67	64	13	21
7	4	8	11	−59	9	7·831	*N*	162	78	10	13
8	4	8	352	−58	7	7·898	*N*	79	79	7	10
9	2	4	2	−56	7	3·984	*N*	123	78	7	10
10	3	6	355	−56	11	5·876	*N*	96	78	11	15
11	4	8	349	−64	6	7·909	*N*	33	82	8	10
12	3	6	37	−71	20	5·571	*N*	236	66	31	36
13	3	6	6	−59	14	5·779	*N*	148	81	16	22
14	2	4	63	−59	30	3·706	*N*	217	45	33	45
15	2	4	357	−57	14	3·931	*N*	101	80	15	21
16	3	6	352	−64	9	5·915	*N*	44	84	11	14
17	3	6	346	−70	11	5·869	*N*	347	80	16	19
18	3	6	348	−59	12	5·832	*N*	62	78	14	19
19	2	4	329	−70	8	3·978	*N*	355	69	12	14
20	2	4	318	−38	12	3·915	*N*	48	47	9	15
21	3	5	323	−43	11	5·867	*N*	48	53	8	14
22	3	6	149	+75	8	5·935	*R*	335	68	13	14

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Table 3

Mean poles for the Chon-Aike Formation obtained using Fisher's statistics giving unit weight to the virtual geomagnetic poles computed for each sampling site. SAJ_1 is the new palaeomagnetic pole of Middle Jurassic age proposed for South America.

Name	Number of Sampling sites	Polarity of nrm	Palaeomagnetic poles Long. E (°)	Lat. S (°)	R	α_{95} (°)
P. Deseado	22	$N-R$	42	84	20·649	8
Camarones (Creer *et al.* 1972)	17	$N-R$	231	82	15·441	12
Ea. La Reconquista	15	N	193	69	13·609	13
Chon-Aike Formation (Pole SAJ_1)	54	$N-R$	197	85	48·749	6

at 84° S, 56° E for the Chon-Aike Formation. In this paper the mean directions of the cleaned remanence for the 22 sampling sites in the Puerto Deseado Area are presented in (Table 2). They were computed giving unit weight to the directions of the cleaned remanence of the disks cut from samples collected at each site. The virtual geomagnetic pole for each sampling site is also given in Table 2 and plotted in Fig. 3. The mean of these 22 virtual geomagnetic poles located at 84° S, 42° E (Table 3 and Fig. 4) is fairly close to the position of the pole computed by Valencio & Vilas (1970) from the directions of cleaned remanence of the 66 useful samples.

3. Discussion

The positions of the mean virtual geomagnetic poles obtained for the Chon-Aike Formation exposures from Puerto Deseado, Camarones, and Estancia La Reconquista (Table 3) are not well grouped, Fig. 4. This may indicate that either the non-dipolar components of the geomagnetic field are not thoroughly eliminated in the mean of the virtual geomagnetic poles for each area or the occurence of unrecognized post-emplacement tectonic movements between the areas. The mean of these three poles is: 83° S, 198° E with $\alpha_{95} = 21°$. The positions of the mean virtual geomagnetic poles obtained from the 22 sites from Puerto Deseado (Table 3), from the normal polarity sites from Camarones (Creer *et al.* 1972), and from the younger units from Estancia La Reconquista (2.2) fall into a compact group (Fig. 4). This suggests that no relative movements have occurred between these areas. The mean poles computed for the reverse polarity sites from Camarones (Creer *et al.* 1972) and the older units from Estancia La Reconquista (2.2) are out of this population. That suggests that the positions of these mean poles are affected by either the secular variation or a reversal of the geomagnetic field. The mean of the last mentioned five poles is 78° S, 198° E with $\alpha_{95} = 24°$.

If we assume that each one of the sampling sites studied at the Estancia La Reconquista, Puerto Deseado and Camarones areas represents an independent geological unit, the most probable position for the palaeomagnetic pole for the Chon-Aike Formation is that obtained as the mean of all the virtual geomagnetic poles computed for these sampling sites (Table 3), and plotted in Fig. 3. The reasons given before suggest that it represents the most reliable position for the Chon-Aike palaeomagnetic pole. Therefore, we propose as the most representative position of the palaeomagnetic pole for the Middle Jurassic Chon-Aike Formation that computed as the mean of the 54 virtual geomagnetic poles from the three studied areas. This Middle Jurassic pole for South America has been named SAJ_1 in Table 3. Note that the position of this pole is close to that proposed by Creer *et al.* (1972) for the Chon-Aike lavas from the Camarones area.

The Permo-Triassic and Mesozoic palaeomagnetic poles for South America which

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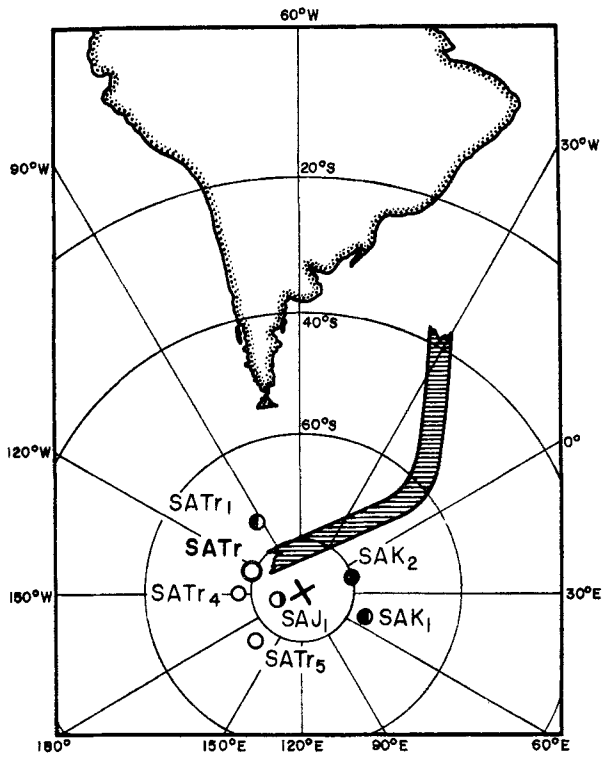


FIG. 5. Late Palaeozoic section of the South American polar wandering curve and Permo-Triassic and Mesozoic palaeomagnetic poles.

satisfy a minimum criterion of reliability (Valencio & Vilas 1972) are shown in Fig. 5. The mean of the 'time group' formed by the Permo-Triassic ($SATr_1$) and the Triassic ($SATr_4$, $SATr_5$) palaeomagnetic poles is: 78·6° S, 321° E with $\alpha_{95} = 16°$ (SATr, Valencio and Vilas, 1972). SAK_1 and SAK_2 are the palaeomagnetic poles for the Lower Cretaceous Serra Geral (Creer 1962) and Vulcanitas Cerro Colorado Formations (Valencio 1972), respectively. The Middle Jurassic palaeomagnetic pole SAJ_1, seems to form a population with the poles $SATr_1$, $SATr_4$ and $SATr_5$. That means that all these poles would form an 'age group' whose mean is 80° S, 216° E with $\alpha_{95} = 10°$, and suggests that South America had a 'quasi-static' period between the Permo-Triassic and the Middle Jurassic. SAK_1 and SAK_2 form a group independent of the Permo-Triassic-Middle Jurassic 'age group' suggesting a polar shift between the Middle Jurassic and the Lower Cretaceous for South America.

When Africa is rotated in relation to South America in order to join their Atlantic coast lines using a pole of rotation at 329° E 44° N with angle of rotation = 57° (Smith & Hallam 1970), the mean of the group of the African Mesozoic palaeomagnetic poles, Af_{6-14}, at 64° S 81° E with $\alpha_{95} = 5°$ (Valencio & Vilas 1969) is rotated to 76° S 217° E in a co-ordinate system referred to South America. Therefore the position of this group is only 4° away from that of the Permo Triassic–Middle Jurassic 'age group' for South America. This suggests that South America and Africa would have been joined until the Middle Jurassic and that the fragmentation of this block would have been simultaneous with the Middle Jurassic–Lower Cretaceous polar shift of South America. However note that this interpretation is based on the position of only one South American palaeomagnetic pole of Middle Jurassic age. Although the position of this pole is fairly well defined it is considered

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desirable to carry out palaeomagnetic studies on other formations of this age with the object of establishing the Jurassic pole position for South America beyond doubt.

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